Cross-chart identification for the weighted-Proj genus-2 model

Formal proof packet

27 October 2026

This is a formal-proof packet, not a new target experiment or cost comparison. The editable diagram and machine-readable gate table are `cross-chart-inverse.svg` and `cross-chart-inverse.csv` beside this PDF.

Scope and checked theorem

For every field K and coefficients $a_3, a_2, a_1, a_0 \in K$, the existing weight-$(1,3,1)$ sextic model has two basic-open charts. In `WeightedProjCrossInjective.lean`, the theorem `cross` identifies affine- and infinity-chart points in the separately glued scheme whenever their images in the actual weighted Proj coincide. The theorem `gluedToProj_isIso` then proves that the canonical glued-to-Proj morphism is a scheme isomorphism. Neither statement requires smoothness or squarefreeness.

Checked proof dependency

Basic-open chart isomorphisms $\longrightarrow$ opposite-open preimages

$\longrightarrow$ cross-chart point equality $\longrightarrow$ global scheme isomorphism

Still open

global divisor/Pic^0 equation compiled Rust semantics

Verification and limits

The direct compile of `WeightedProjCrossInjective.lean` exited zero with Lean 4.34.1 and Mathlib `d13f23b`. The Markdown companion gives the exact command. Both theorem axiom audits listed only `propext`, `Classical.choice`, and `Quot.sound`; neither listed `sorryAx`. Source SHA-256:

`d59ee7093bca3842cada43b055590331b11393f378106776a00ee9b6ddc28f8b`.

Formal gate	Status	Scope
Opposite-open preimages	checked	arbitrary field/coefficient
Cross-chart point identification	checked	arbitrary field/coefficient
Glued-to-Proj scheme isomorphism	checked	arbitrary field/coefficient
Global divisor/Pic^0 class equation	open	genus-2 law
Compiled Rust exceptional branches	open	runtime law

The theorem does not alter any frozen 1,260-target row, operation counter, matched-rho control, or cost ratio. The measured comparison remains in the four-follow-up report. The class equation and source-semantics bridge remain independent obligations before treating the complete inversion-free law as proved end to end.